

CS 736

Medical Image Computing (MIC)

Kernel Methods

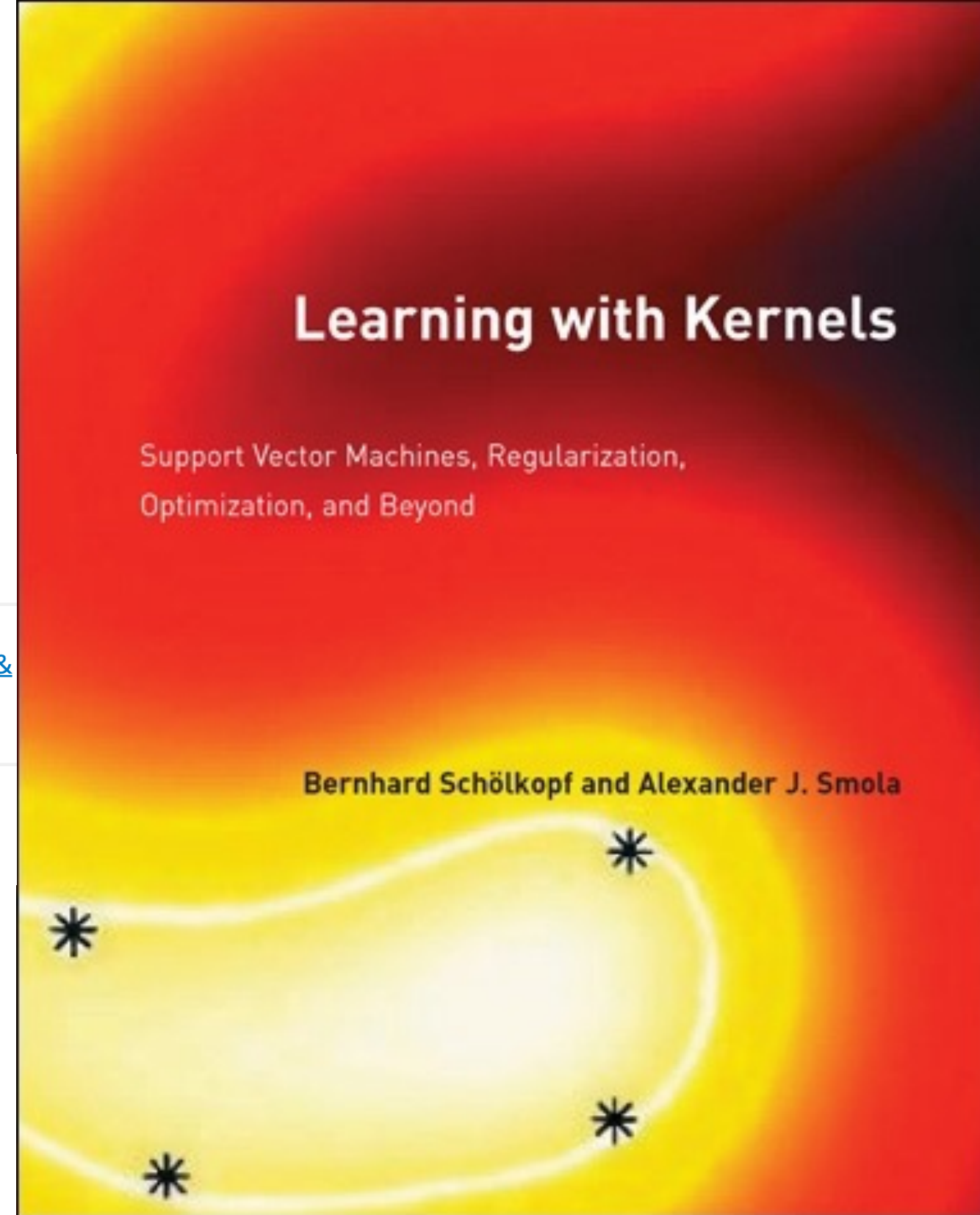
Suyash P. Awate

References

Nonlinear component analysis as a kernel eigenvalue problem

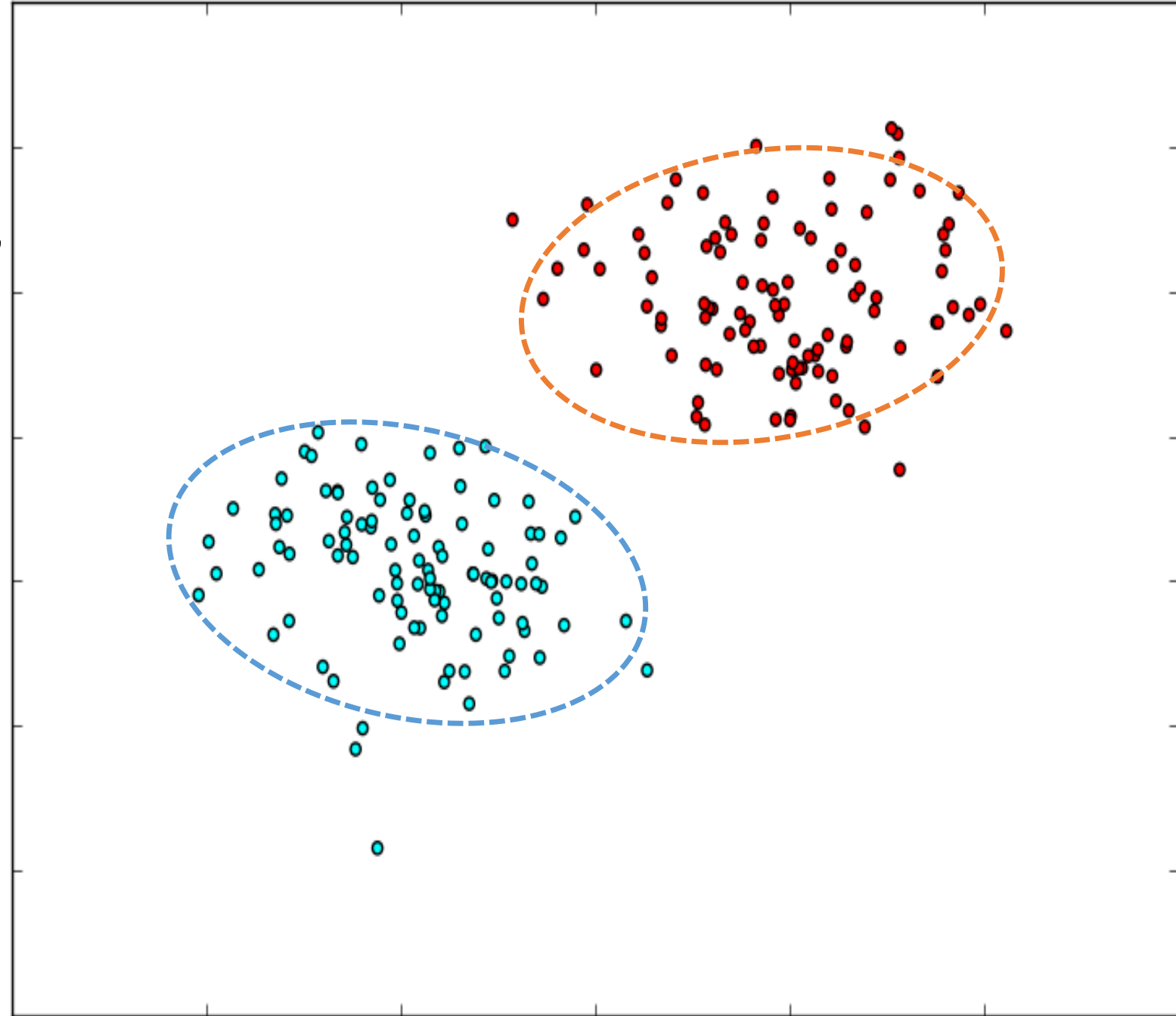
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Neural Computation, Volume 10, Issue 5 • July 1, 1998 • pp 1299–1319 • <https://doi.org/10.1162/089976698300017467>



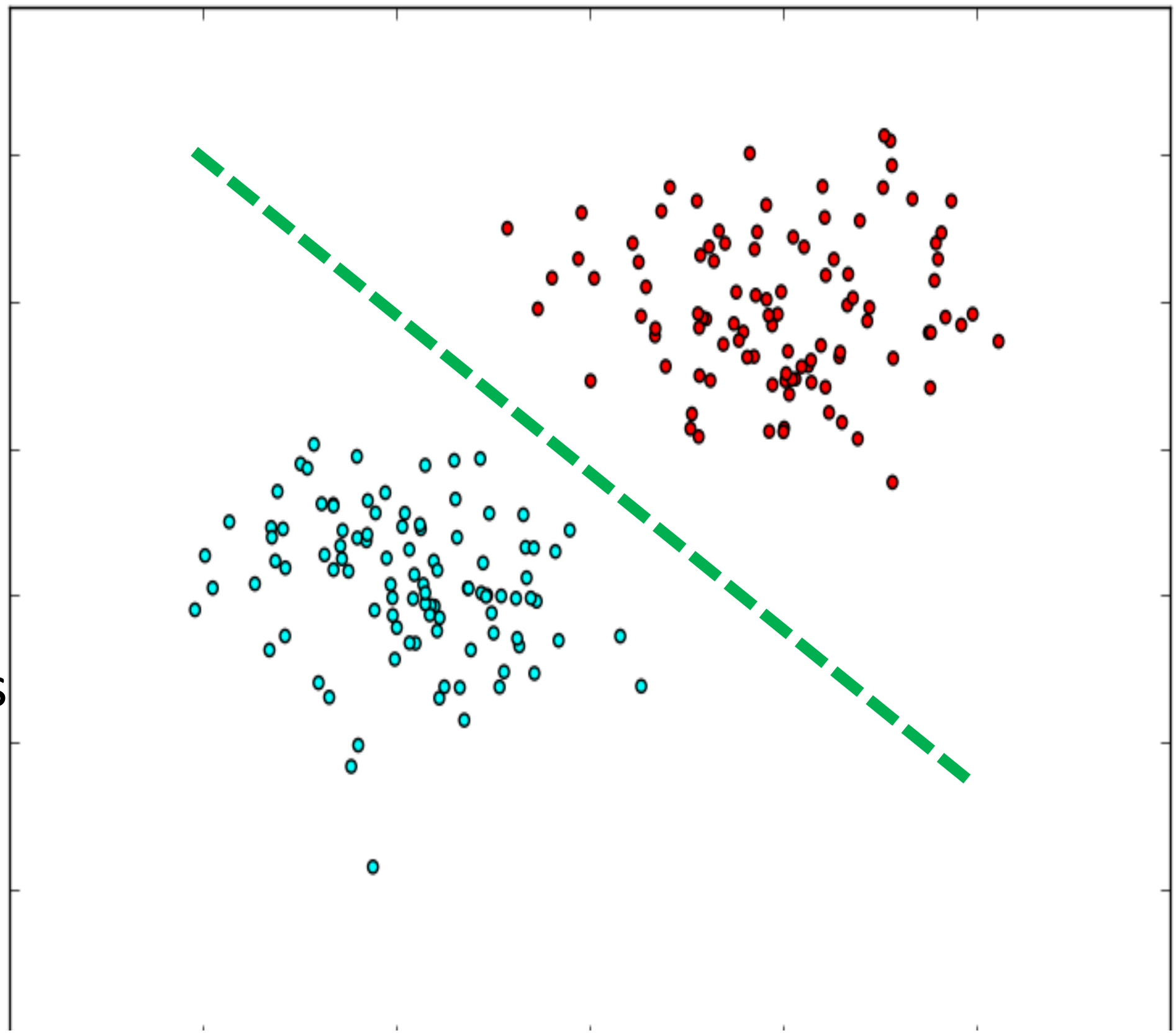
Modeling Data

- **Generative models**
- Example:
When data in **ellipsoidal** clusters, we can model each cluster distribution using a **multivariate Gaussian**
 - 1) Can be used for **sampling**
 - 2) Can be used for **classification**
 - For a test point, assign point to cluster that gives largest value of probability density function



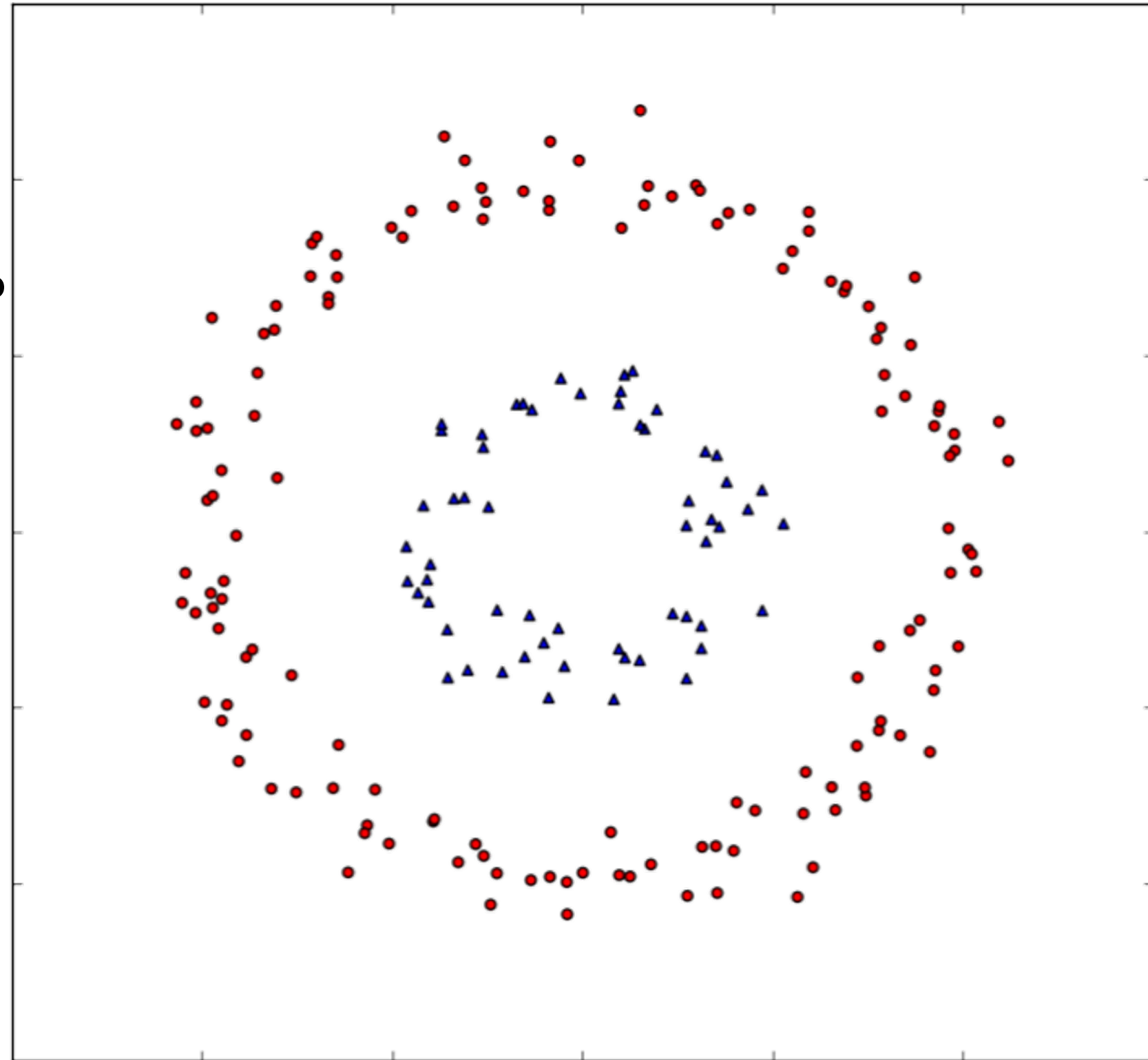
Modeling Data

- **Discriminative models**
(for classification)
- Example:
When data is **linearly separable**,
use a hyperplane
to separate classes,
as in
support vector machines



Modeling Data

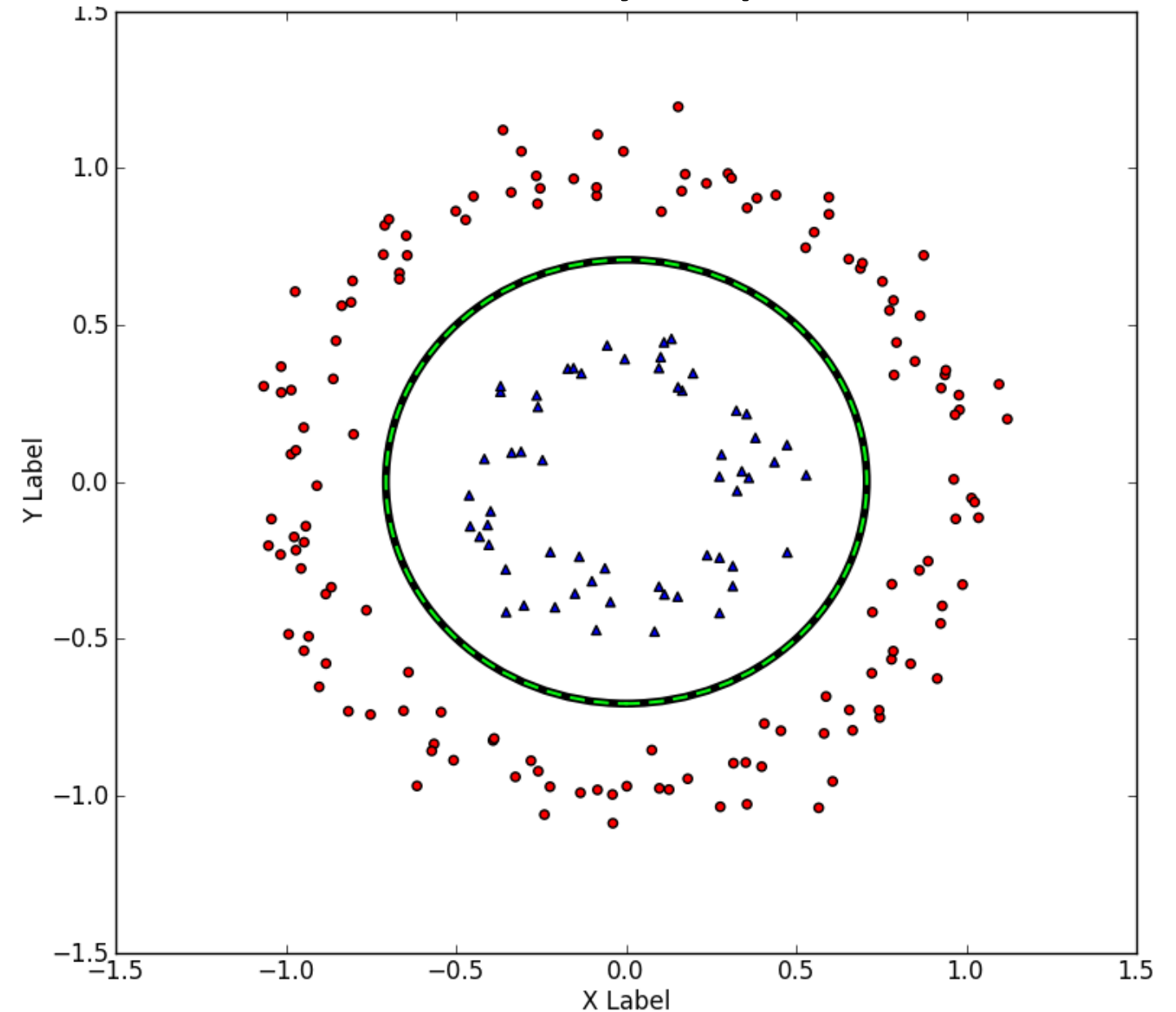
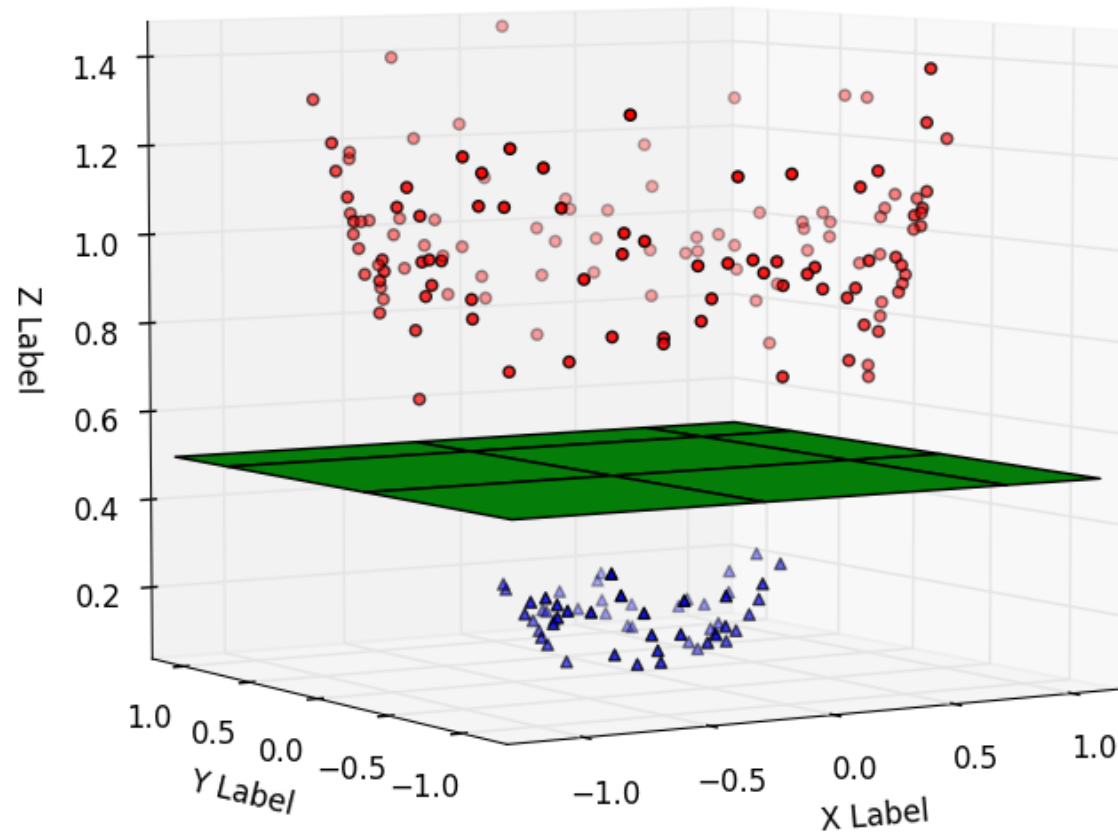
- **Discriminative** modeling
(for classification)
 - What to do when classes **not linearly** separable ? but when decision boundary is a complex nonlinear one
- **Generative** modeling
 - What to do when a data cluster **doesn't** exhibit **ellipsoidal** distributions ? but when distribution is more complex



Modeling Data

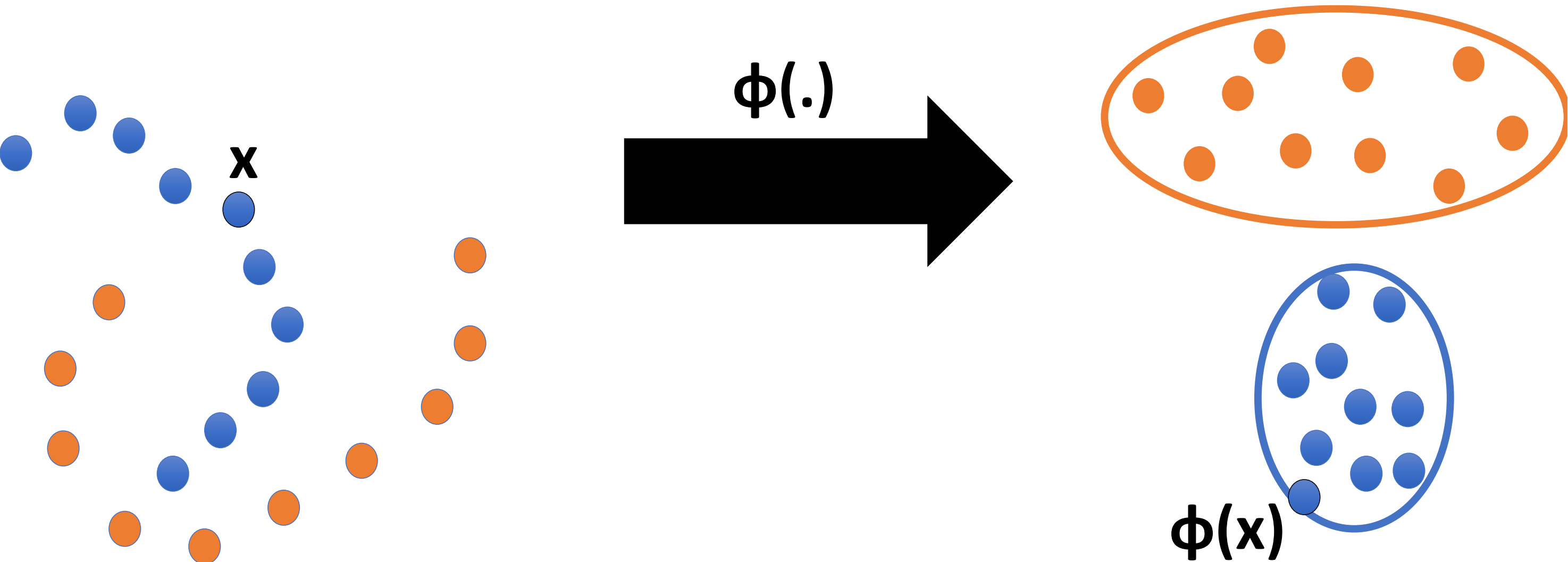
- Idea: nonlinearly mapping data to a different space
- e.g., map $[x, y] \rightarrow [x, y, z:=x^2+y^2]$ makes data linearly separable

Data in R^3 (separable w/ hyperplane)



Modeling Data

- Idea: nonlinearly mapping data to a different space
- e.g., design a (nonlinear) mapping so that, in the mapped space, cluster distributions can be modeled by Gaussians



Kernel Trick

- Mapped space can be very high dimensional
 - Datum = x
 - Mapped datum = $\phi(x)$
- Mapping function $\phi(\cdot)$ can be very nonlinear/complex
- Computation and storage costs associated with $\phi(\cdot)$ prevent us from working with mapped data $\phi(x)$ explicitly
- Instead, we use a 'kernel' function $k(\cdot, \cdot)$
 - Kernel gives inner product in mapped space: $k(x, y) = \langle \phi(x), \phi(y) \rangle$
 - Don't store $\phi(x)$ at all
 - Use $k(x, y)$ for all computations
- Identity map $\phi(\cdot) \rightarrow k(x, y) = \langle x, y \rangle$

Kernel Trick

- In practice, we choose/design a kernel first !
- Our goal: to design a mapping function $\phi(\cdot)$ equivalent to a kernel $k(\cdot, \cdot)$
- Some theorems describe conditions on kernel for existence of an equivalent mapping function
 - e.g., $k(\mathbf{x}, \mathbf{y}) = \exp(-a \|\mathbf{x} - \mathbf{y}\|_2^2)$ where 'a' is a free parameter
 - Gaussian kernel; **radial basis function (RBF) kernel**
 - Mapped space has infinite dimension

$$\exp\left(-\frac{1}{2}\|\mathbf{x} - \mathbf{x}'\|^2\right) = \sum_{j=0}^{\infty} \frac{(\mathbf{x}^\top \mathbf{x}')^j}{j!} \exp\left(-\frac{1}{2}\|\mathbf{x}\|^2\right) \exp\left(-\frac{1}{2}\|\mathbf{x}'\|^2\right)$$

- e.g., for degree $j=2$, points $\mathbf{x}=[x_1, x_2]$, $\mathbf{y}=[y_1, y_2]$ in 2D:

$$(\mathbf{x} \cdot \mathbf{y})^2 = (x_1^2, x_1 x_2, x_2 x_1, x_2^2)(y_1^2, y_1 y_2, y_2 y_1, y_2^2)^\top$$

Kernel Trick

- Mapped space:
 - Can be finite dimensional, i.e., a Euclidean space
 - Can be **infinite dimensional**, i.e., Hilbert space of **functions**
- Hilbert space
 - **Vector space** having the structure of an **inner product**
 - What kind of space is called a vector space ?
 - What kinds of functions can be called inner products ?

Kernel Trick

- **Vector space defines** operations of **addition** and **scaling** on elements of this space (called “vectors”)

Axiom

Meaning

Associativity of addition

$$\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$$

Commutativity of addition

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$$

Identity element of addition

There exists an element $\mathbf{0} \in V$, called the *zero vector*, such that $\mathbf{v} + \mathbf{0} = \mathbf{v}$ for all $\mathbf{v} \in V$.

Inverse elements of addition

For every $\mathbf{v} \in V$, there exists an element $-\mathbf{v} \in V$, called the *additive inverse* of $\mathbf{v}$, such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$.

Compatibility of scalar multiplication with field multiplication

$$a(b\mathbf{v}) = (ab)\mathbf{v} \text{ [nb 2]}$$

Identity element of scalar multiplication

$1\mathbf{v} = \mathbf{v}$, where 1 denotes the *multiplicative identity* in F .

Distributivity of scalar multiplication with respect to vector addition

$$a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$$

Distributivity of scalar multiplication with respect to field addition

$$(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$$

Kernel Trick

- **Inner-product** definition must satisfy 3 conditions:

1) Conjugate **symmetry**: $\langle f, g \rangle = \langle g, f \rangle^*$

2) **Linearity** in first argument: $\langle a_1 f_1 + a_2 f_2, g \rangle = a_1 \langle f_1, g \rangle + a_2 \langle f_2, g \rangle$

3) **Positive definite**: $\langle f, f \rangle \geq 0$ with equality iff $f = 0$

- A zero function is a function that is almost-everywhere zero

- Inner product allows us to measure:

- Angle between two vectors
- Distances/lengths, through induced norm

Kernel Principal Component Analysis

- **David Hilbert (1862-1943)**
- German mathematician, among most influential of his time
 - Invariant theory, calculus of variations, commutative algebra, algebraic number theory, foundations of geometry, spectral theory
- Student (non-PhD): **John von Neumann**
 - Hungarian and American mathematician, physicist, computer scientist and engineer
- Hilbert's problems
 - 23 math problems published in 1900



Kernel Trick

1) “Reproducing kernel” (RK) function

Definition 2 (Reproducing Kernel). Let $\mathcal{H}$ be a Hilbert space of functions $f : \mathcal{X} \rightarrow \mathbb{R}$ defined on a non-empty set $\mathcal{X}$. A function $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is called a reproducing kernel of $\mathcal{H}$ if it satisfies

- (1) $\forall x \in \mathcal{X}, k_x = k(x, \cdot) \in \mathcal{H}$,
- (2) $\forall x \in \mathcal{X}, \forall f \in \mathcal{H}, \langle f, k_x \rangle_{\mathcal{H}} = f(x)$ (the reproducing property).

In particular, for any $x, y \in \mathcal{X}$, $k(x, y) = \langle k_x, k_y \rangle_{\mathcal{H}} = \langle k_y, k_x \rangle_{\mathcal{H}} = k(y, x)$.

- “Reproducing kernel” is representor of evaluation (or, evaluation functional)

• Reproducing kernel Hilbert space (RKHS)

Theorem 3. A Hilbert space $\mathcal{H}$ is a reproducing kernel Hilbert space if and only if it has a reproducing kernel.

Theorem 4. If it exists, reproducing kernel is unique. Equivalently, a reproducing kernel Hilbert space uniquely determines its reproducing kernel.

- Evaluation functionals are bounded, continuous

Kernel Trick

2) “Kernel” (K) function

Definition 3 (Kernel). A function $k : \mathcal{X} \times \mathcal{X} \longrightarrow \mathbb{R}$ is called a kernel on $\mathcal{X}$ if there exists a Hilbert space (not necessarily a reproducing kernel Hilbert space) $\mathcal{F}$ and a map $\Phi : \mathcal{X} \longrightarrow \mathcal{F}$ such that $k(x, y) = \langle \Phi(x), \Phi(y) \rangle_{\mathcal{F}}$.

- Corollary: Every “reproducing kernel” function is a “kernel” function; where $\phi(x) := k(x, \cdot)$

3) “Positive definite” (PD) function

Definition 4. A symmetric function $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is positive definite if $\forall n \in \mathbb{N}$, $\forall (a_1, \dots, a_n) \in \mathbb{R}^n$ and $\forall (x_1, \dots, x_n) \in \mathcal{X}^n$,

$$\sum_{i=1}^n \sum_{j=1}^n a_i a_j k(x_i, x_j) \geq 0.$$

The function $k(\cdot, \cdot)$ is strictly positive definite if for mutually distinct x_i , the equality holds only when all the a_i are zero.

- Corollary: Every “kernel” function is a “positive definite” function; because, in Hilbert space $\mathcal{F}$, inner-product of $(\sum_i a_i \phi(x_i))$ with itself = squared-norm of $(\sum_i a_i \phi(x_i)) \geq 0$

Kernel Trick

4) Moore-Aronszajn Theorem

Theorem 5 (Moore-Aronszajn). *Let $k : \mathcal{X} \times \mathcal{X} \longrightarrow \mathbb{R}$ be positive definite. Then there is a unique reproducing kernel Hilbert space with reproducing kernel k . In particular, the span of the reproducing kernel k is dense in the resulting reproducing kernel Hilbert space.*

- Proof: Construct H (with inner product) as “completion” of H_0 , where

Let the space $\mathcal{H}_0 = \text{span}\{k(x, \cdot) : x \in \mathcal{X}\}$ be endowed with the inner product

$$\langle f, g \rangle_{\mathcal{H}_0} = \sum_{i=1}^n \sum_{j=1}^m \alpha_i \beta_j k(x_i, y_j), \quad \text{where } f = \sum_{i=1}^n \alpha_i k(x_i, \cdot) \text{ and } g = \sum_{j=1}^m \beta_j k(y_j, \cdot)$$

- Our designed **inner product** is:
 - **Symmetric**: $\langle f, g \rangle = \langle g, f \rangle$, because $k(.,.)$ is symmetric
 - **Linear**: $\langle f+e, g \rangle = \langle f, g \rangle + \langle e, g \rangle$ and $\langle af, g \rangle = a \langle f, g \rangle$, by construction
 - **Positive definite**: because $k(.,.)$ is positive definite and so $\langle f, f \rangle \geq 0$ and $\langle f, f \rangle = 0$ iff $f=0$
- Then, reproducing property holds: $\langle k(x, .), k(y, .) \rangle = k(x, y)$

REPRODUCING



HILBERT
SPACE

Kernel Principal Component Analysis

- **E H Moore** (1862–1932)
 - American mathematician
 - PhD advisor, at Yale, was Hubert Newton, who also advised Gibbs
 - First Head of math dept. at U Chicago
 - Stayed Head until death
 - Co-invented Moore-Penrose pseudo-inverse (1920)
- **Nachman Aronszajn** (1907–1980)
 - Polish-American mathematician
 - Main work: “systematically developed the concept of RKHS” in field of mathematical analysis
 - Two PhD degrees:
 - (1) In Poland (1930)
 - (2) In France (1935): advisor was Maurice Frechet



Kernel Mapping $\leftrightarrow$ Mixture Model for Density Estimation

- Mean of mapped data: $\Phi_0 = 1/n \sum_{i=1}^n \Phi(\mathbf{x}_i)$
- Distance (squared) between mapped datum $\phi(\mathbf{z})$ and mean ϕ_0 is:

$$\|\Phi(\mathbf{z}) - \Phi_0\|^2 = k(\mathbf{z}, \mathbf{z}) - \frac{2}{n} \sum_{i=1}^n k(\mathbf{z}, \mathbf{x}_i) + \frac{1}{n^2} \sum_{i,j=1}^n k(\mathbf{x}_i, \mathbf{x}_j)$$

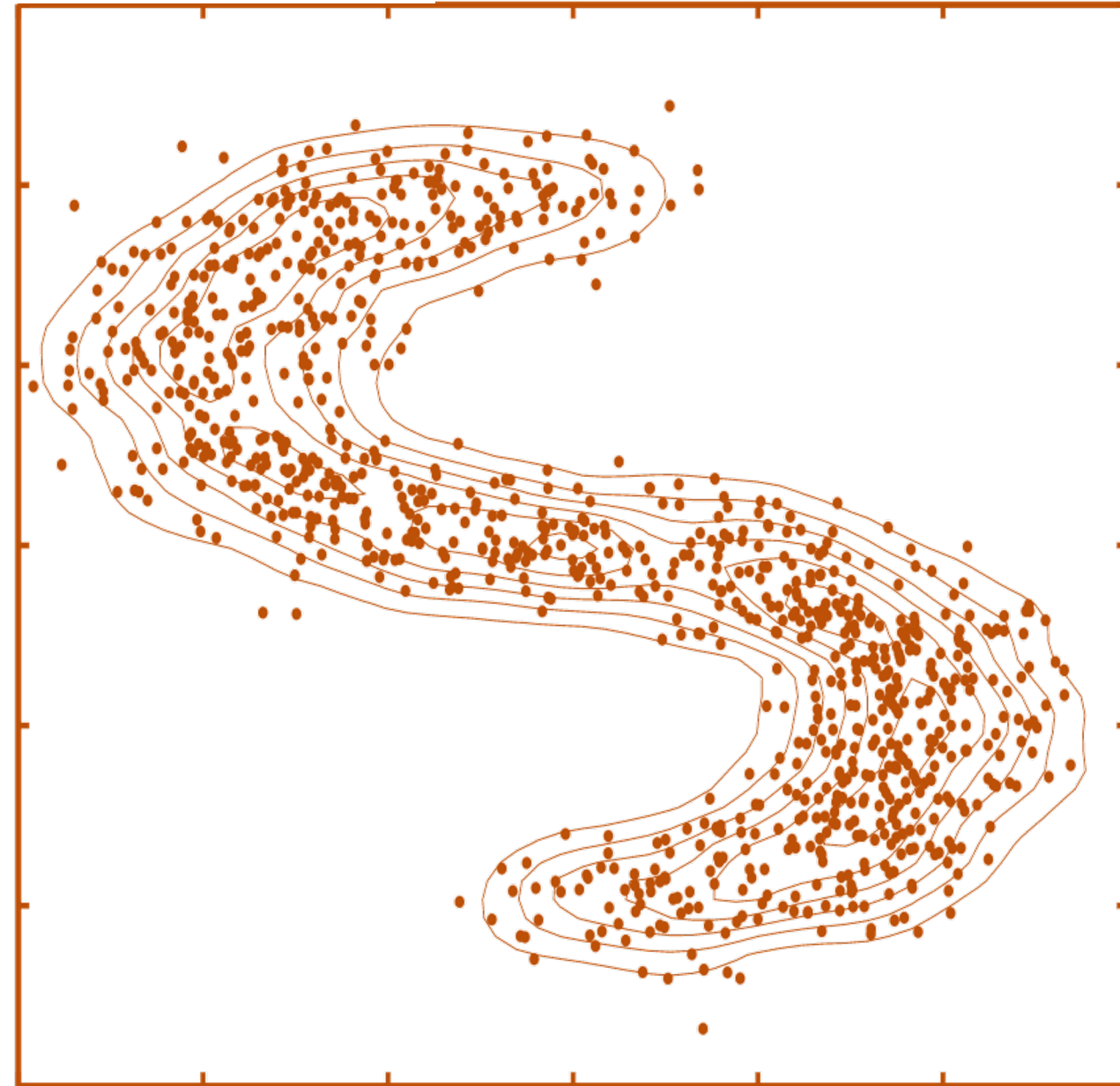
- 3rd term = constant (independent of $\mathbf{z}$) $\rightarrow$ ignore
- 1st term = constant for RBF kernel $\rightarrow$ ignore
- 2nd term equivalent to using density estimation using RBF (Gaussian) mixture model
 - Larger PDF of $\phi(\mathbf{z})$ (via RBF mixture-model) in input space $\rightarrow$ closer is $\phi(\mathbf{z})$ to mean ϕ_0 in (RKHS) feature space

Kernel Mapping $\leftrightarrow$ Mixture Model for Density Estimation

- Contours of squared-distance function $\|\Phi(\mathbf{z}) - \Phi_0\|^2$

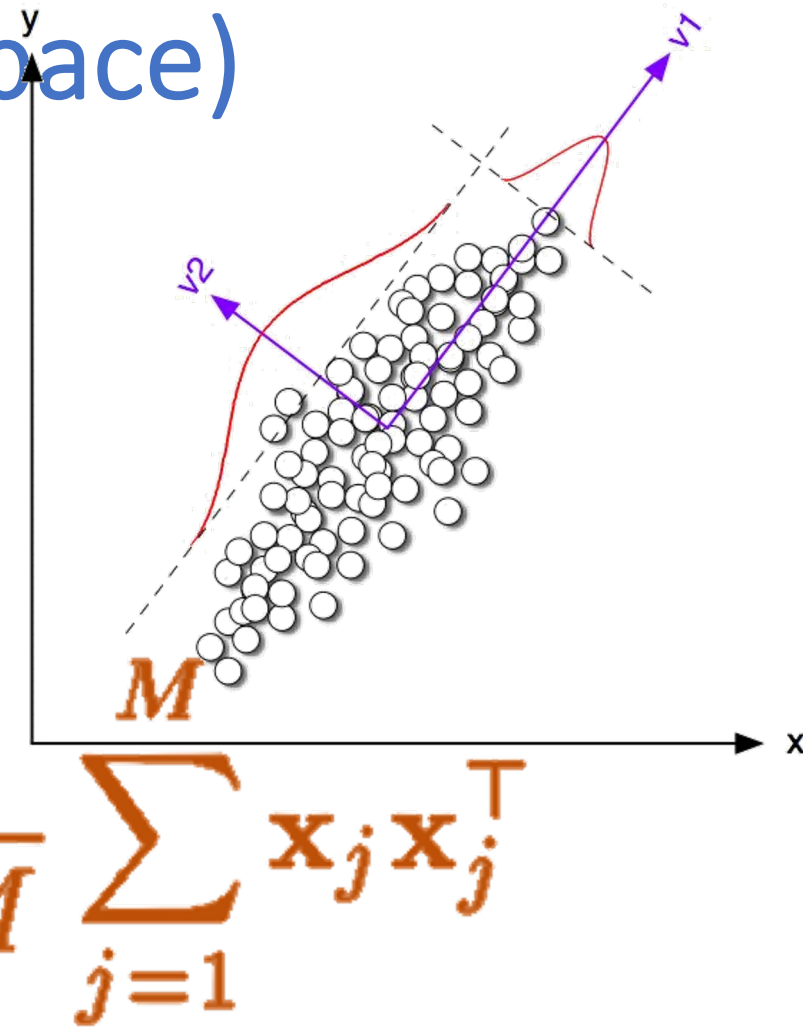
are same as contours of $\sum_{i=1}^n k(\mathbf{z}, \mathbf{x}_i)$

- How do these contours look like ?
 - Distance-function contours adapt to non-ellipsoidal distribution of data
 - Without kernel-based nonlinear mapping (i.e., with identity kernel $\phi(\mathbf{z}) := \mathbf{z}$) (i.e., with 1 Gaussian in input space), contours are circular, not adapted to data
 - Covariance model will make them ellipsoidal, but still not adapted to data
- When using kernels, what does covariance model in RKHS give us ?



Principal Component Analysis (Euclidean space)

- Data sample = $\{ \mathbf{x}_j : j=1, \dots, M \}$
- Mean estimate = sample mean
- Now, considered **centered** data
 - Shift in coordinate frame s.t. sample mean is at origin
- Covariance matrix estimate
(for **centered** data) = sample covariance = $\mathbf{C} = \frac{1}{M} \sum_{j=1}^M \mathbf{x}_j \mathbf{x}_j^T$
- Eigenvectors “v” of covariance C
→ directions/modes of variation

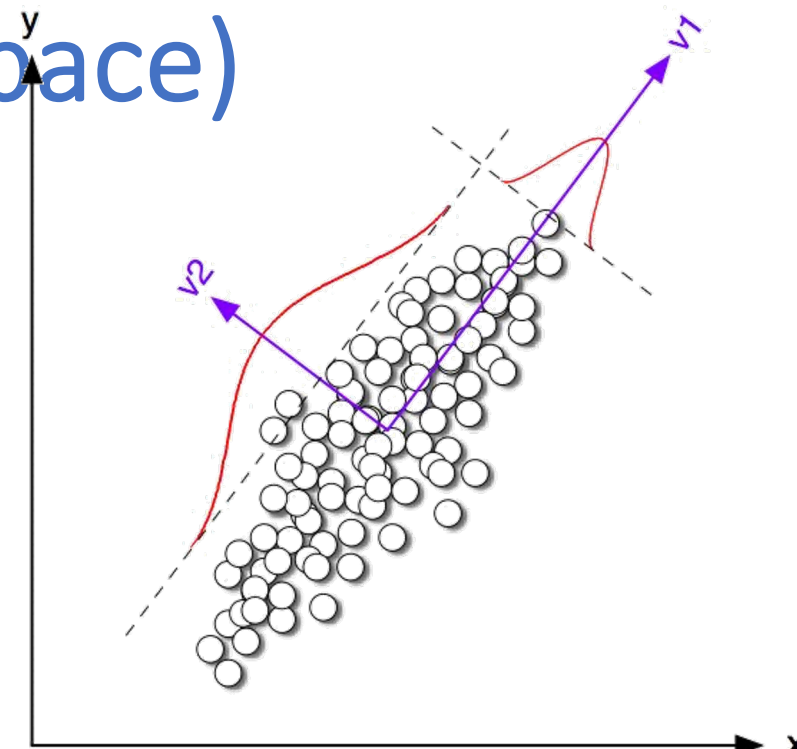


- $\lambda \mathbf{v} = \mathbf{C} \mathbf{v} \quad \Rightarrow \quad \mathbf{v} = \frac{1}{\lambda} \mathbf{C} \mathbf{v} = \frac{1}{M} \sum_{j=1}^M (\mathbf{x}_j \cdot \mathbf{v}) \mathbf{x}_j$

- Eigenvector (with positive eigenvalue) must lie in span of (**centered**) data

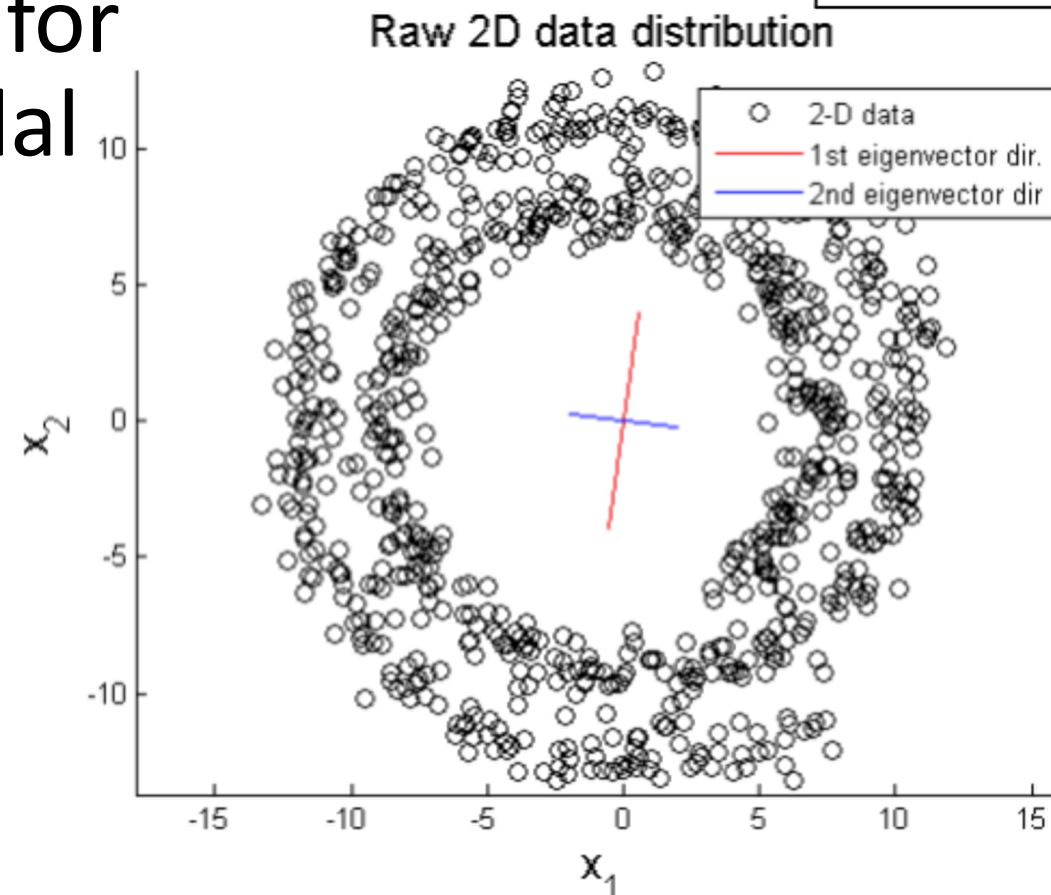
Principal Component Analysis (Euclidean space)

- PCA works great for Gaussian-distributed data



- But PCA doesn't work very well for a data cluster that isn't ellipsoidal

- e.g., what if data has the following distribution ?

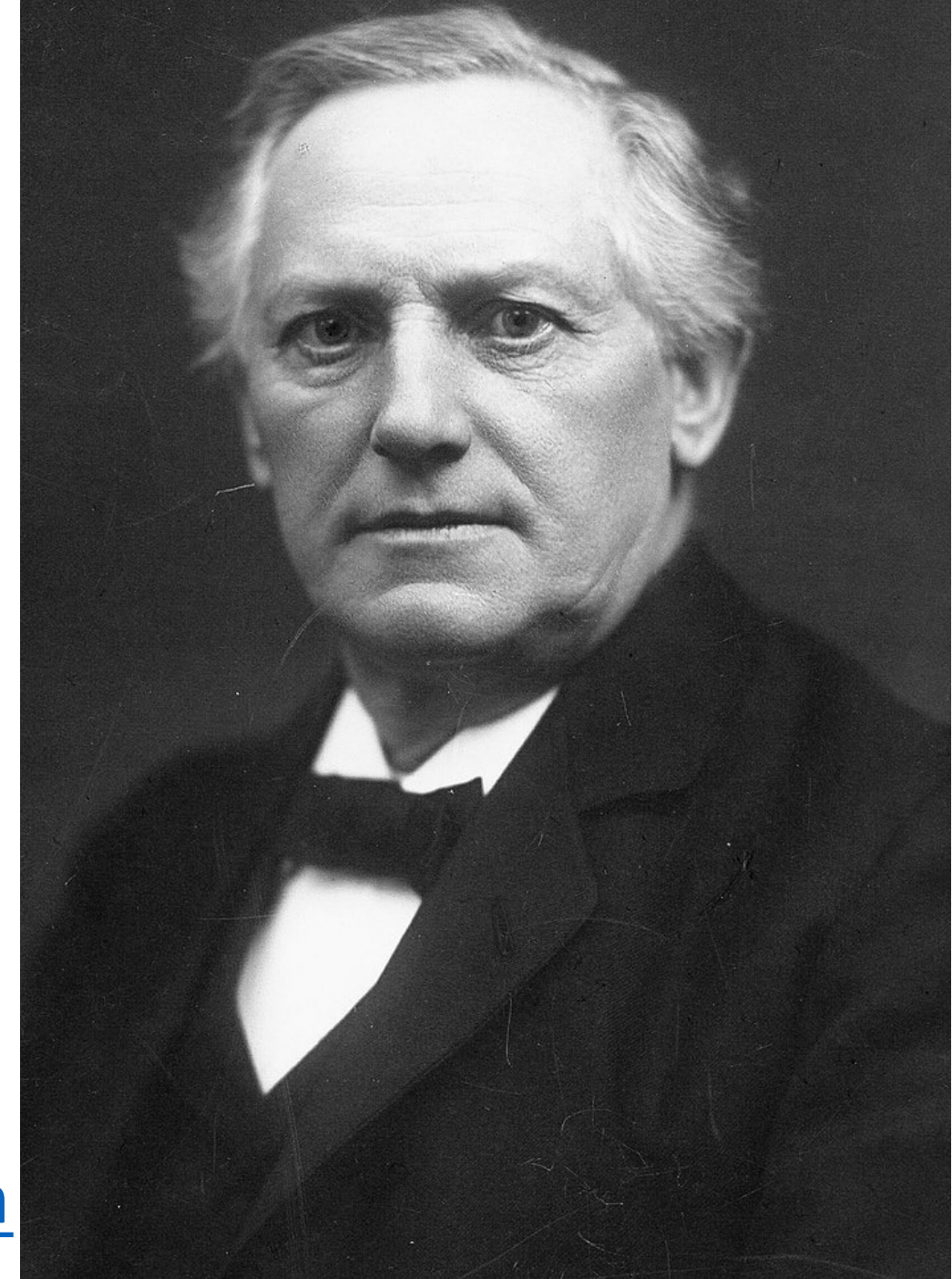


Kernel Principal Component Analysis

- For **centered** data **in RKHS**, covariance “matrix” is: $\bar{C} = \frac{1}{M} \sum_{j=1}^M \Phi(\mathbf{x}_j) \Phi(\mathbf{x}_j)^\top$
- We can **represent** eigenvector $\mathbf{V} = \sum_{i=1}^M \alpha_i \Phi(\mathbf{x}_i)$
- For eigenvector $\mathbf{V}$: $\lambda \mathbf{V} = \bar{C} \mathbf{V}$
 - Inner product of LHS and RHS with a $\Phi(\mathbf{x}_k)$: $\lambda(\Phi(\mathbf{x}_k) \cdot \mathbf{V}) = (\Phi(\mathbf{x}_k) \cdot \bar{C} \mathbf{V})$
- Substituting forms of $\mathbf{V}$ and $\bar{C}$ in this equation, we get, for each k , ...
- These systems of equations can be simplified to $M \lambda \mathbf{G} \alpha = \mathbf{G}^2 \alpha$
where $\mathbf{G} = \text{Gram matrix}$ for centered data s.t. $G_{ij} := \langle \phi(\mathbf{x}_i), \phi(\mathbf{x}_j) \rangle$
- Solve $M \lambda \alpha = \mathbf{G} \alpha$
 - All solutions α (complete set of eigenvectors of $\mathbf{G}$) satisfy **previous equation**
- Eigenvectors of $\mathbf{G}$ give vectors α that represent $\mathbf{V}$ (rescale to norm 1)
- λ = eigenvalues of covariance matrix $\mathbf{C} = (1/M)$ eigenvalues of $\mathbf{G}$

Kernel Principal Component Analysis

- Gram matrix, named after **Jorgen Gram**
- Matrix of inner products of a set of vectors
- Danish mathematician: 1850–1916
- Risk analysis for insurance company
- “His brilliance and scientific training together with his **practical skills** made his contributions to **pure and applied mathematics** very significant.”
 - mathshistory.st-andrews.ac.uk/Biographies/Gram
- Also invented Gram–Schmidt orthogonalization algorithm
 - But people found later that Laplace (1749–1827) already knew it



Kernel Principal Component Analysis

- **Linear modes** of variation 'v' in **mapped space** (RKHS) implicitly produce **nonlinear modes** in **input space** (because of nonlinear $\phi(\cdot)$)
 - Call "feature" = $\langle \phi(x), v \rangle$
 - Visualize contours of "feature" values over all points 'x' in input space

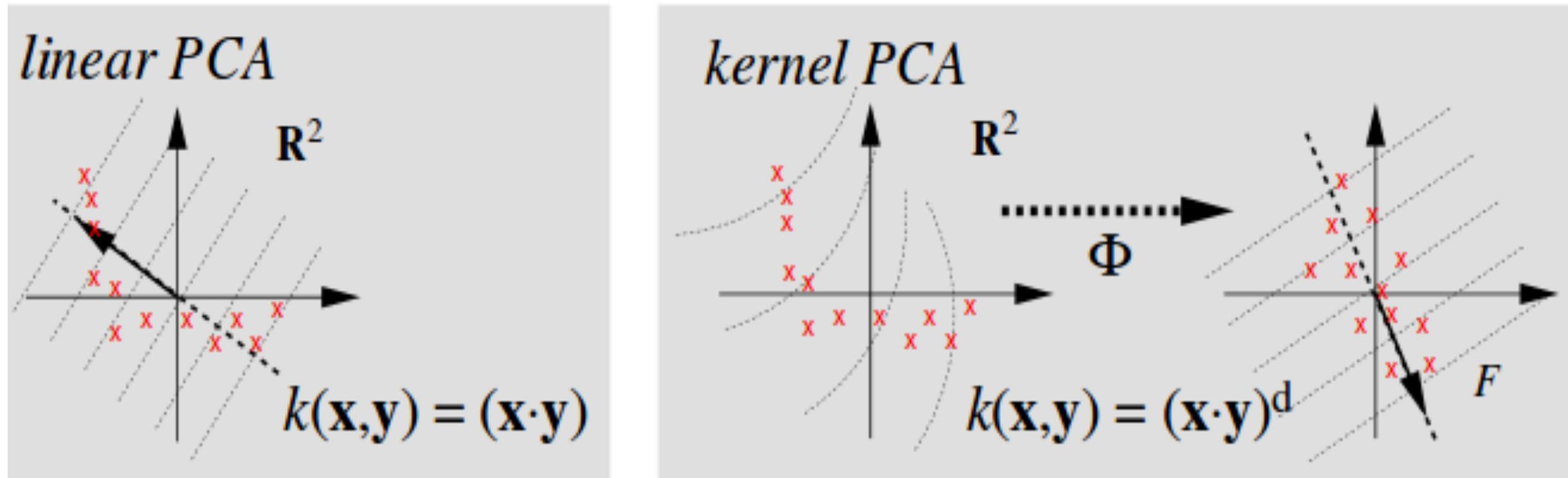


Fig. 1. Basic idea of kernel PCA: by using a nonlinear kernel function k instead of the standard dot product, we implicitly perform PCA in a possibly high-dimensional space F which is nonlinearly related to input space. The dotted lines are contour lines of constant feature value.

Kernel Principal Component

- Plot contours of distance from **subspace** selected by few principal eigenvectors
- Can be better than GMM, because GMM only looks at distance from **mean** in RKHS
- Kernel choice matters

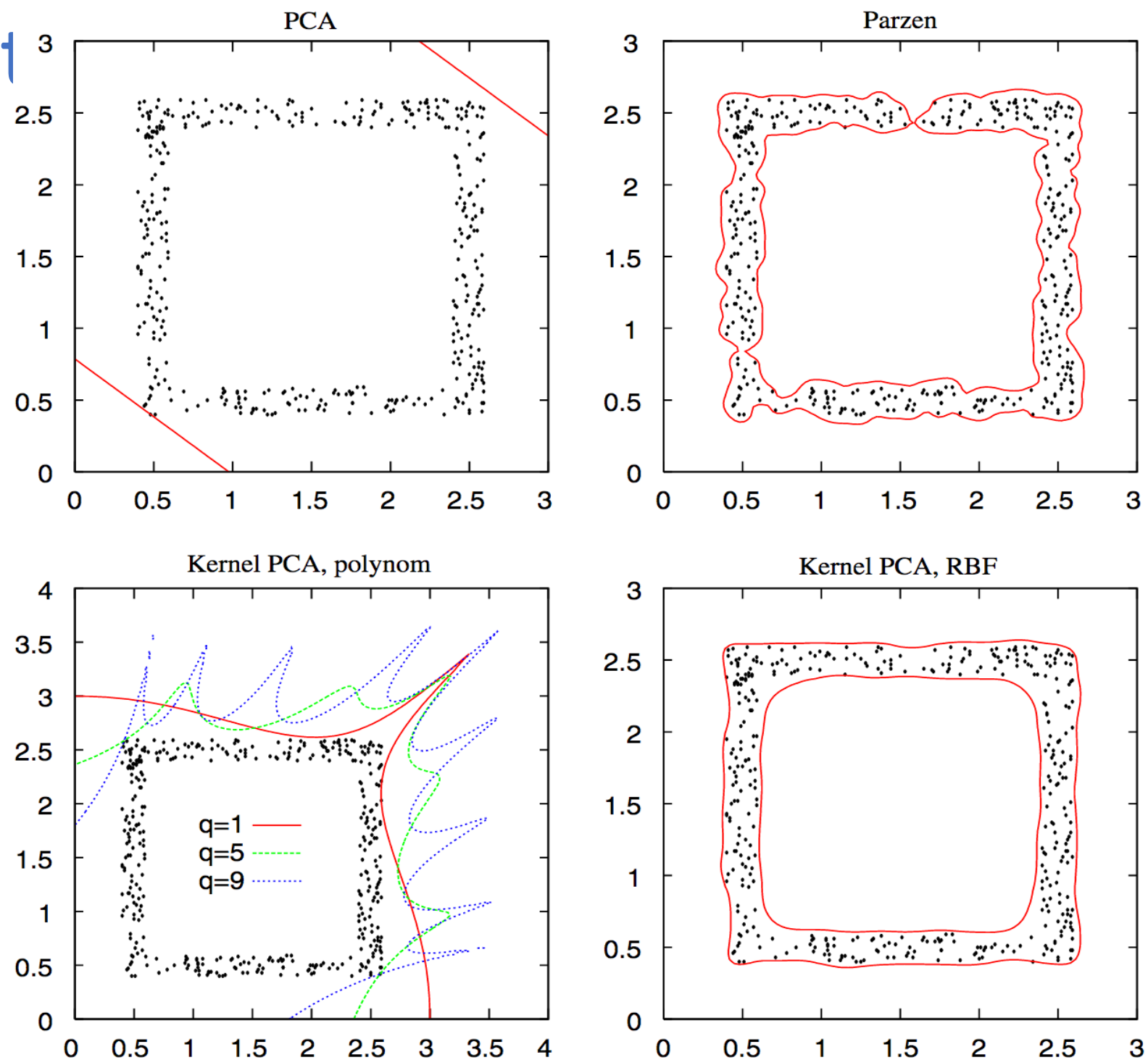
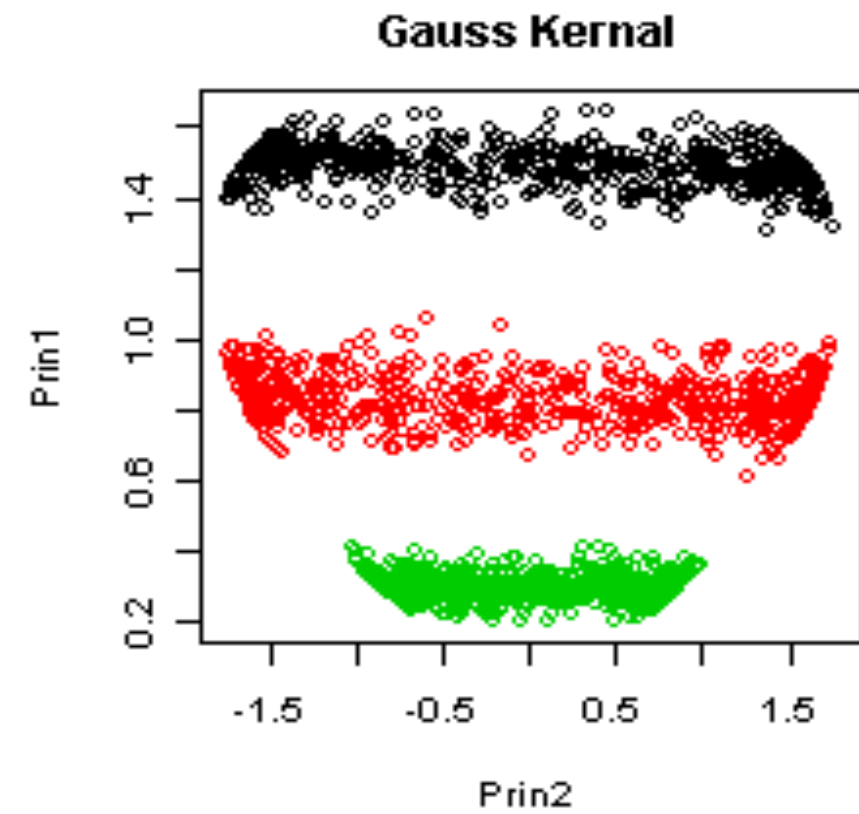
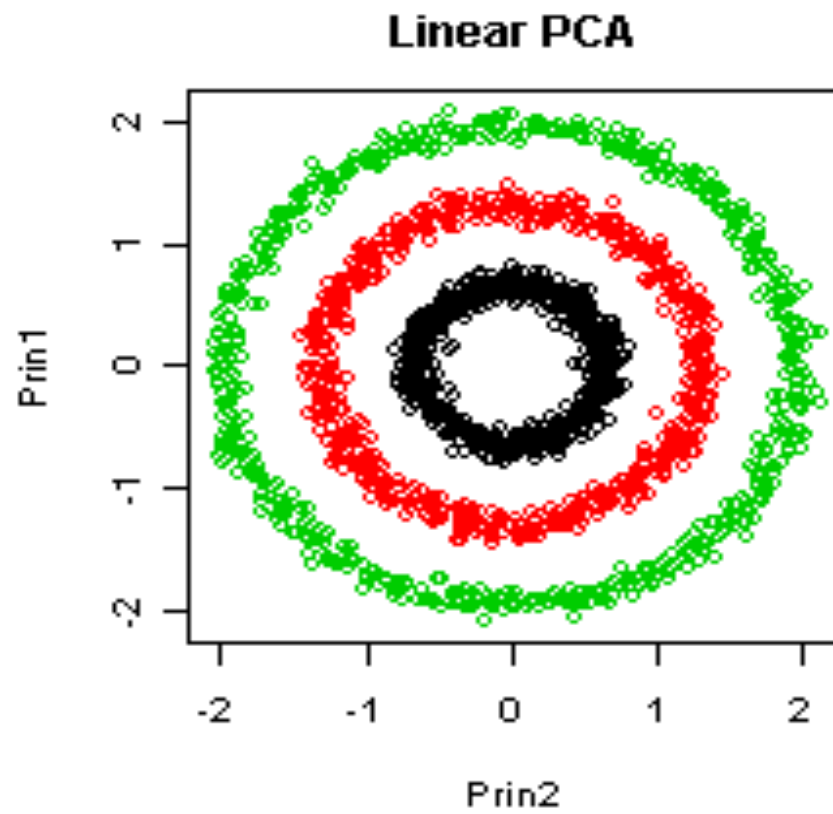
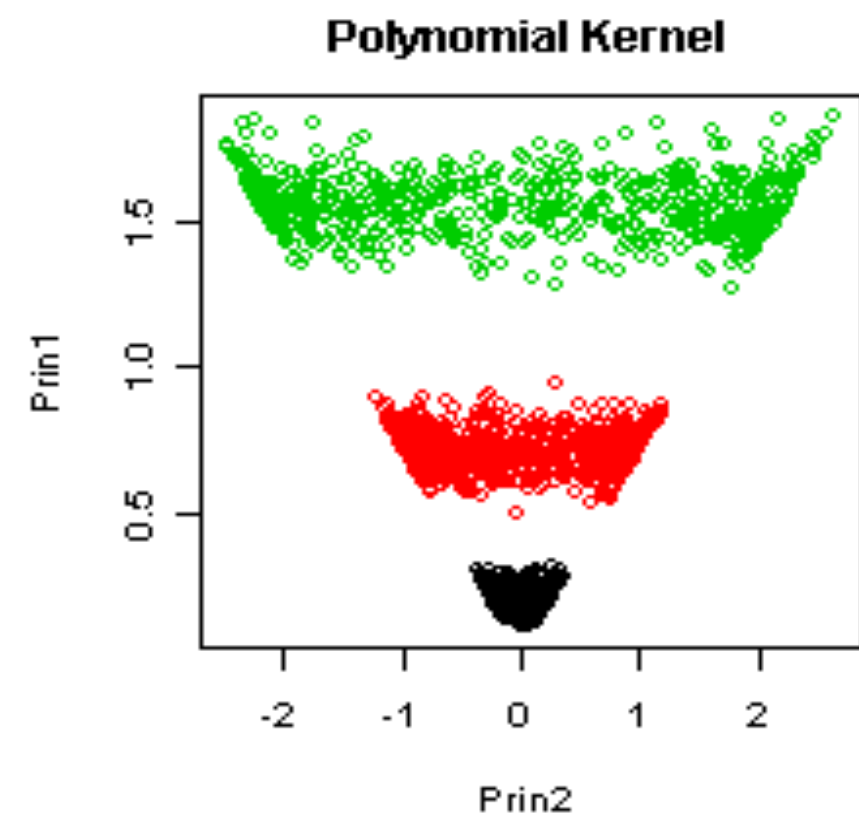
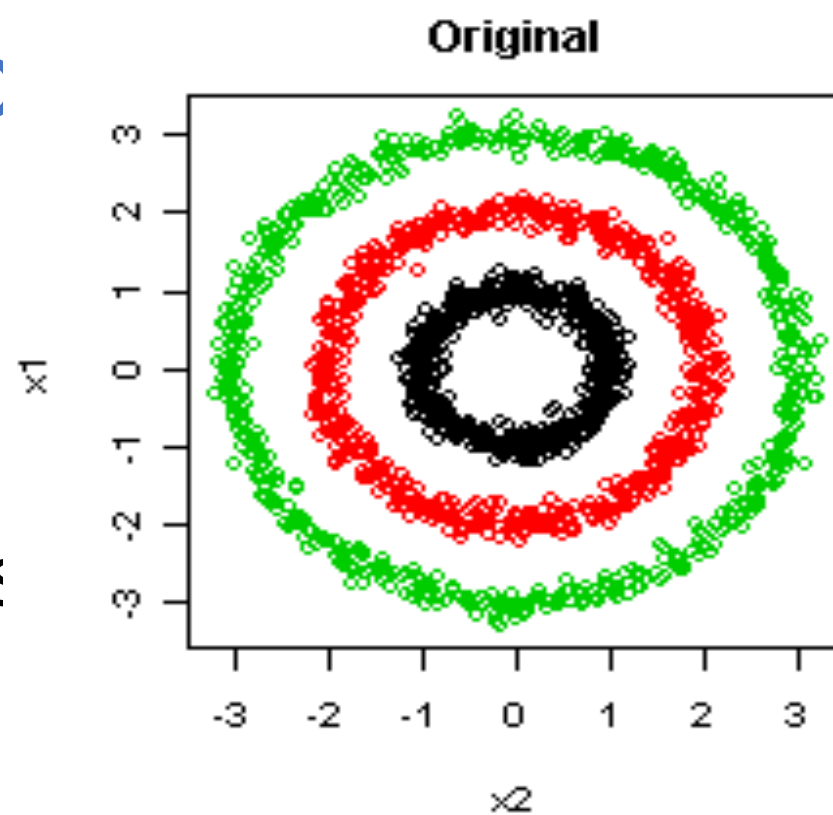


Fig. 2. Decision boundaries for various methods: (top left) PCA reconstruction error with $q = 1$ eigenvector, (top right) Parzen window density estimator with $\sigma = 0.05$, (bottom left) reconstruction error in $\mathcal{F}$ with polynomial kernel $k(\mathbf{x}_i, \mathbf{x}_j) = (\mathbf{x}_i \cdot \mathbf{x}_j)^{10}$ and various q values, (bottom right) reconstruction error in $\mathcal{F}$ with Gaussian kernel using $\sigma = 0.4$ and $q = 40$.

Kernel Principal Compc

- More examples
- Projecting mapped data onto top 2 principal components in mapped space
- KPCA separates clusters clearly better than PCA
- Nonlinear dimensionality reduction



Parameter Tuning

- “**Grad-student descent**”
- = Grad student fiddles with parameters until it works
[Ryan Adams, Harvard]



Kernel Methods in Medical Image Computing

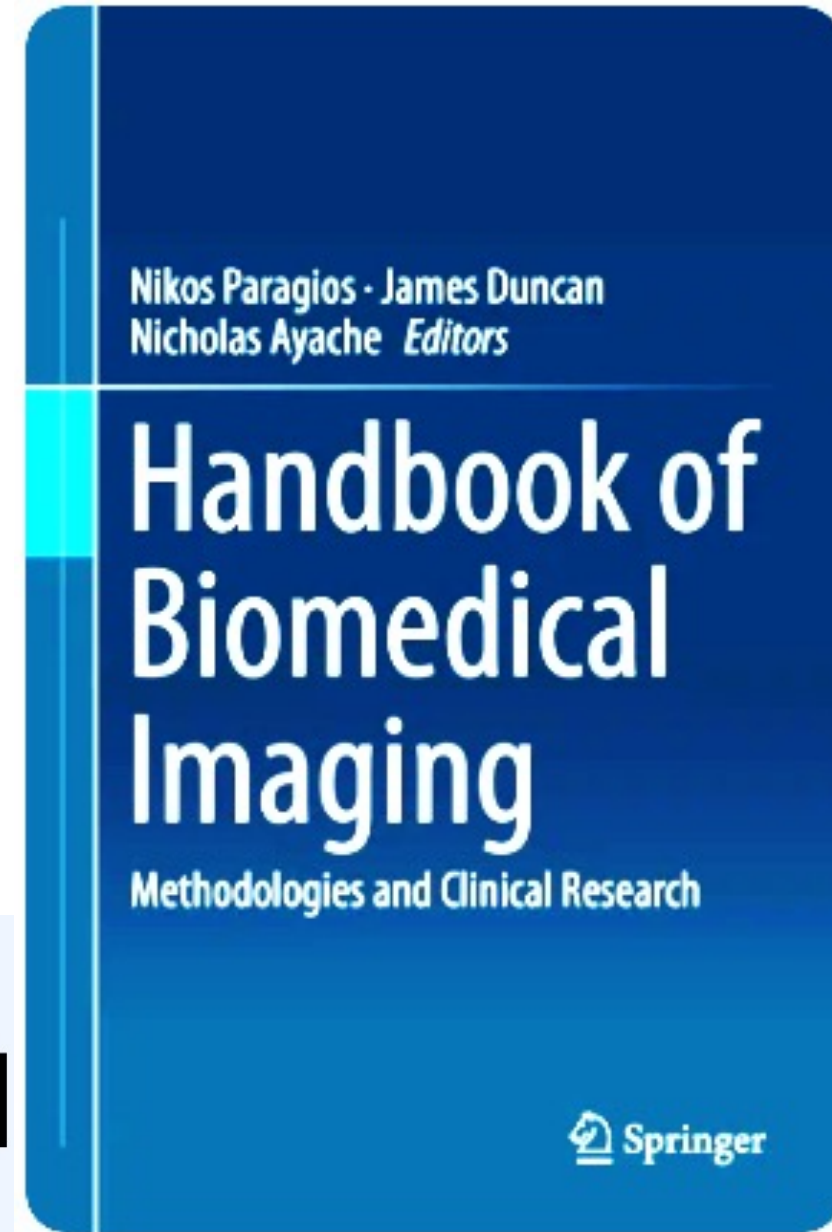
- Kernel PCA
 - For dimensionality reduction
 - For image segmentation and clustering: Kernel k-means / FCM / GMM
 - For [statistical shape analysis](#)
- Kernel ridge regression ([tutorial](#))
 - For image restoration: [denoising](#) , [motion correction](#)
 - For image classification: in [fMRI](#) , [brain connectivity](#)
 - For image [registration](#)

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Kernel Methods in Medical Imaging

Chapter

pp 63–81 | [Cite this chapter](#)



Kernel Ridge Regression

- Ridge regression
- Independent variables (covariates) { vectors $\mathbf{x}_i$ in $\mathbb{R}^d$ } _{$i=1,\dots,n$} (centered)
- Dependent (= response) variables { scalars y_i } _{$i=1,\dots,n$} (centered)
- Regressor model: $y = \mathbf{w}^T \mathbf{x}$, with parameters $\mathbf{w}$

- Objective function

$$\frac{1}{2} \sum_i (y_i - \mathbf{w}^T \mathbf{x}_i)^2 + \frac{1}{2} \lambda \|\mathbf{w}\|^2$$

- At optimal $\mathbf{w}$,
gradient w.r.t. $\mathbf{w}$ is 0

$$\sum_i (y_i - \mathbf{w}^T \mathbf{x}_i) \mathbf{x}_i = \lambda \mathbf{w}$$

- Solve linear system
 - Note: $(\mathbf{w}^T \mathbf{x}_i) \mathbf{x}_i = \mathbf{x}_i (\mathbf{x}_i^T \mathbf{w})$

$$\mathbf{w} = \left(\lambda \mathbf{I} + \sum_i \mathbf{x}_i \mathbf{x}_i^T \right)^{-1} \left(\sum_j y_j \mathbf{x}_j \right)$$

Kernel Ridge Regression

- **Kernel** trick: **Replace** $\mathbf{x}_i$ by $\phi(\mathbf{x}_i)$ (assume centered)

$$\mathbf{w} = \left(\lambda \mathbf{I} + \sum_i \mathbf{x}_i \mathbf{x}_i^T \right)^{-1} \left(\sum_j y_j \mathbf{x}_j \right)$$

- Consider “matrix” Φ (capital phi) with i-th column as $\phi(\mathbf{x}_i)$
- Consider vector $\mathbf{y}$ with j-th component as y_j
- But, for typical (e.g., RBF) kernels, $\Phi\Phi^T$ is high-dimensional (e.g., ∞ -D)
- [Bishop Book, Appendix C](#): “Basic Matrix Identities”

$$\mathbf{w} = (\lambda \mathbf{I}_d + \Phi\Phi^T)^{-1} \Phi\mathbf{y}$$

$$(\mathbf{P}^{-1} + \mathbf{B}^T \mathbf{R}^{-1} \mathbf{B})^{-1} \mathbf{B}^T \mathbf{R}^{-1} = \mathbf{P} \mathbf{B}^T (\mathbf{B} \mathbf{P} \mathbf{B}^T + \mathbf{R})^{-1}.$$

which is easily verified by right multiplying both sides by $(\mathbf{B} \mathbf{P} \mathbf{B}^T + \mathbf{R})$

- Replaces inversion in high-dim (RKHS) to inversion in low dim (data)
- In our case: $\mathbf{P}^{-1} := \lambda \mathbf{I}_d$ and $\mathbf{R} := \mathbf{I}_n$ and $\mathbf{B}^T := \Phi$ (capital phi)

Kernel Ridge Regression

- Then $\mathbf{w} = \Phi(\Phi^T \Phi + \lambda \mathbf{I}_n)^{-1} \mathbf{y}$
- Interpret $\mathbf{w} = \sum_i \alpha_i \Phi(\mathbf{x}_i)$ with $\boldsymbol{\alpha} = (\Phi^T \Phi + \lambda \mathbf{I}_n)^{-1} \mathbf{y}$
- Equivalently $\mathbf{w} = \Phi \boldsymbol{\alpha}$ (with capital phi)
- At new point $\mathbf{x}$, predicted value of dependent variable is:

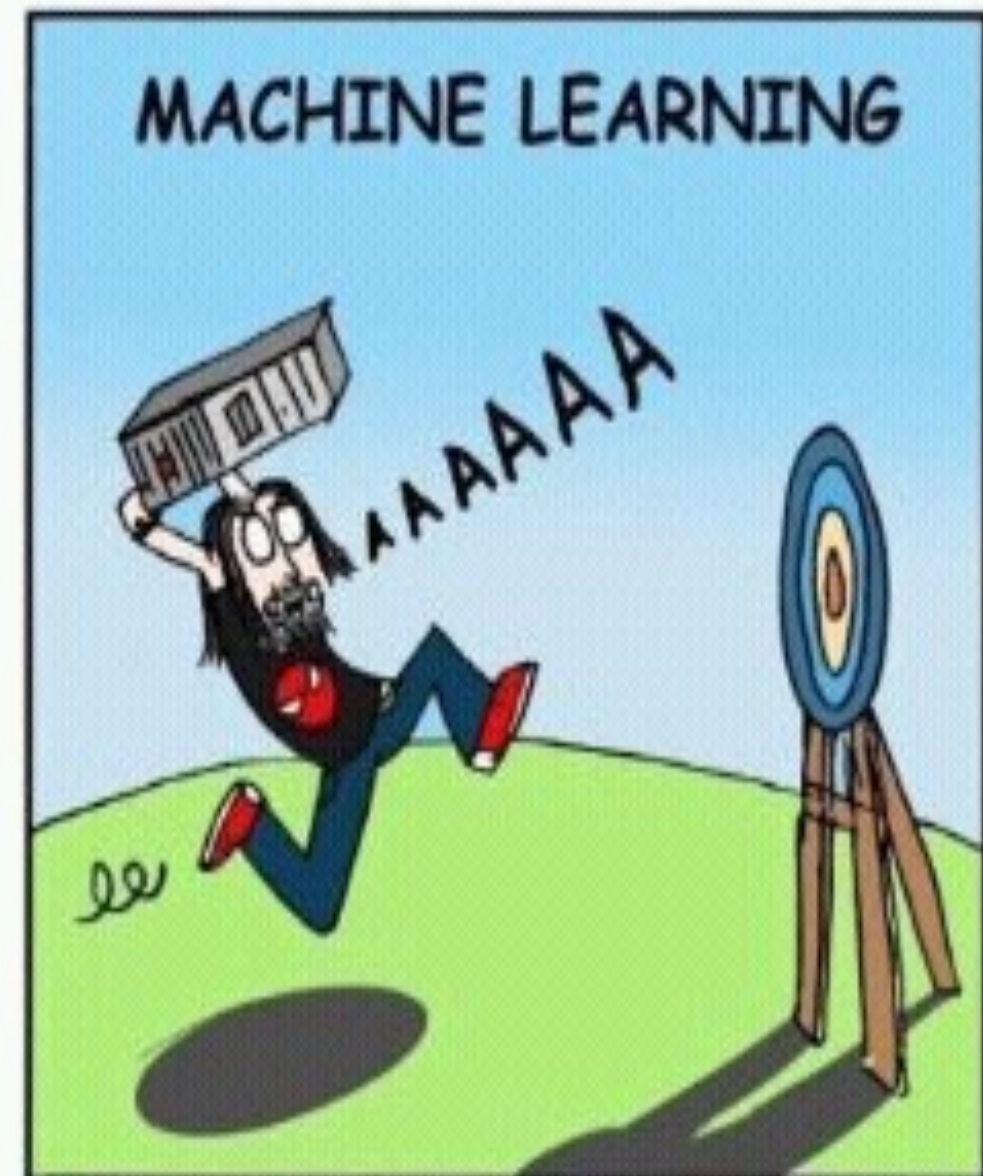
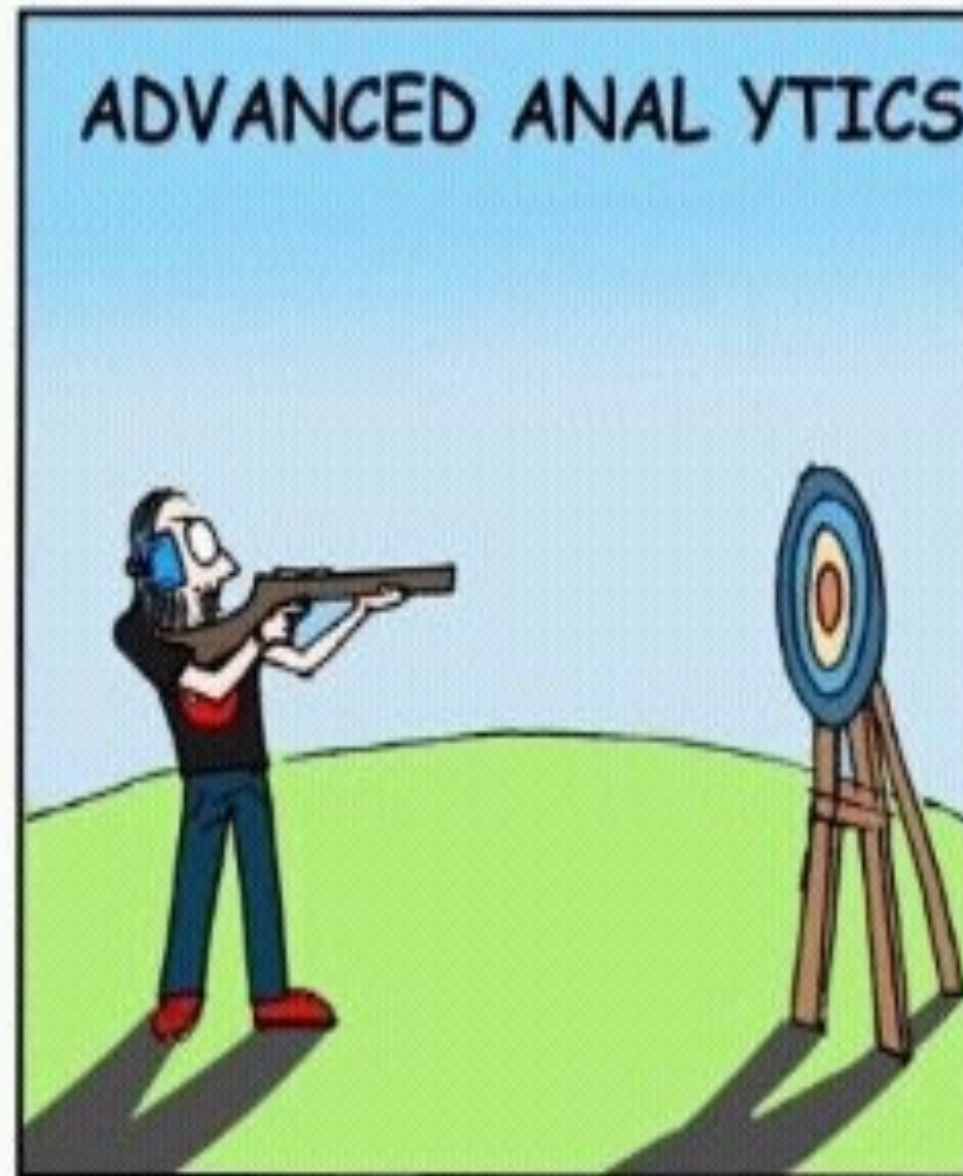
$$\mathbf{w}^T \Phi(\mathbf{x}) = \mathbf{y}^T (\Phi^T \Phi + \lambda \mathbf{I}_n)^{-1} \Phi^T \Phi(\mathbf{x})$$

- because inverse of symmetric matrix $M (= V D V^T)$ is symmetric

$$= \mathbf{y}^T (K + \lambda \mathbf{I}_n)^{-1} \boldsymbol{\kappa}(\mathbf{x})$$

- where $\boldsymbol{\kappa}(\mathbf{x})$ is a vector with components $\langle \phi(\mathbf{x}_i), \phi(\mathbf{x}) \rangle$

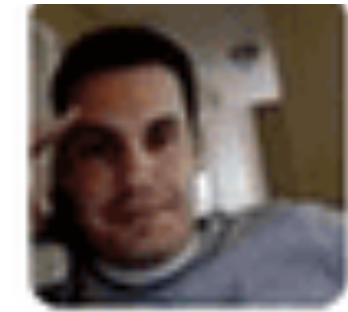
In Lighter Vein



• SM. MAR 2017.

Data Science is Interdisciplinary, Making it Challenging

- Lower bounds for a data scientist



Josh Wills

@josh_wills



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Data Scientist (n.): Person who is better at statistics than any software engineer and better at software engineering than any statistician.

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9:55 AM - 3 May 12

Data Science is Interdisciplinary, Making it Challenging

- Upper bounds for a data scientist
- Will Cukierski of Kaggle:

Data scientist = “Person who is worse at statistics than any statistician and worse at software engineering than any software engineer.”